

Differential Geometry I Midterm

Yosime Flood

January 2025

1 Problem 1

Let (M, ω) be a symplectic manifold and let $H: M \rightarrow \mathbb{R}$ be a smooth function. Denote by X_H the Hamiltonian vector field associated to H via the condition $\omega \circ i_{X_H} = -dH$, where $\omega \circ i_{X_H} := \omega(X_H, \cdot)$.

1.1 (a)

Suppose that $Y \in \mathfrak{X}(M)$ is a smooth vector field with globally defined flow φ_t such that this flow is symplectic: $(\varphi_t)^*(\omega) = \omega$ and leaves H invariant: $H \circ \varphi_t = H$ for all t . Furthermore, suppose that M has trivial first de Rham cohomology. Then Y is the Hamiltonian vector field of a smooth function $F: M \rightarrow \mathbb{R}$ that is preserved under the flow X_H .

Proof. We wish to show that there exists a smooth function $F: M \rightarrow \mathbb{R}$, which is invariant under the flow of X_H , such that

$$\omega \circ i_Y = -dF.$$

We will first prove the existence of a function F and then show that it is preserved under the flow X_H . Since the assumption states that $H_{dR}^1(M) = 0$ it suffices to show that $\omega \circ i_Y$ is a closed 1-form, since then it must also be exact, and we obtain that it is the exterior derivative of some $F \in \Omega^0(M) = C^\infty(M)$. Using Cartan's magic formula, we may compute:

$$d(\omega \circ i_Y) = \mathcal{L}_Y \omega - i_Y d\omega = \mathcal{L}_Y \omega,$$

Where we used the fact that symplectic forms are closed by exercise 14.10. Now the assumption states that ω is constant under the flow of Y , and thus :

$$\mathcal{L}_Y \omega = \left. \frac{d}{dt} (\varphi_t)^*(\omega) \right|_{t=0} = \left. \frac{d}{dt} \omega \right|_{t=0} = 0.$$

Hence, we have shown that $\omega \circ i_Y$ is closed and therefore exact. Let F be a smooth function such that Y is the Hamiltonian vector field of F , we will show that F is constant along flow lines of H . This is equivalent to showing that the

Lie derivative $\mathcal{L}_{X_H} F$ vanishes everywhere. This, however, is also a consequence of Cartan's Magic formula:

$$\begin{aligned}
\mathcal{L}_{X_H} F &= d(i_{X_H} F) + i_{X_H} dF \\
&= i_{X_H} dF \\
&= -\omega(Y, X_H) \\
&= \omega(X_H, Y) \\
&= -dH(Y) \\
&= \mathcal{L}_Y(H).
\end{aligned}$$

Here, we used Cartan's magic formula in the first and last equation. Finally, the last term vanishes since H was assumed to be constant along the flow lines of Y . $\square$

1.2 (b)

Let (x, p_x, y, p_y) be standard coordinates of $\mathbb{R}^4$. We interpret $q := (x, y)$ as position coordinates and $p := (p_x, p_y)$ as momentum coordinates. On the set $\{Y \neq 0\}$, we may use polar coordinates (r, θ) to describe the position coordinates. Furthermore, on this same open set, we may define

$$p_r := \frac{x}{r}p_x + \frac{y}{r}p_y \quad \text{and} \quad p_\theta := -yp_x + xp_y.$$

We claim that the coordinates $(r, p_r, \theta, p_\theta)$ define a darboux chart with respect to the standard symplectic form $w_{\text{st}} := dp_x \wedge dx + dp_y \wedge dy$ on $\mathbb{R}^4$.

Proof. We should briefly explain why the coordinates $(r, p_r, \theta, p_\theta)$ form a chart of $\mathbb{R}^4$ in the first place. We already know that (x, y) is uniquely and smoothly determined by (r, θ) . Furthermore, solving the system of equations defining p_r and p_θ for p_x and p_y , we obtain the momentum coordinates in terms of r, θ, p_r, p_θ , showing that the map $(r, p_r, \theta, p_\theta)$ has smooth inverse and thus it is a chart. We want to show that $w_{\text{st}} = dp_r \wedge dr + dp_\theta \wedge d\theta$ holds on the open set on which these coordinates are defined. To show this, we first observe that w_{st} is actually an exact form. Indeed, if we define $\lambda_{\text{st}} := p_x dx + p_y dy$, then we find:

$$\begin{aligned}
d\lambda_{\text{st}} &= d(p_x dx + p_y dy) \\
&= dp_x \wedge dx + dp_y \wedge dy,
\end{aligned}$$

where we used $d^2 \equiv 0$ to cancel out the other terms. Similarly, we find that

$$dp_r \wedge dr + dp_\theta \wedge d\theta = d(p_r dr + p_\theta d\theta)$$

Thus, to complete the proof, it suffices to show that

$$p_r dr + p_\theta d\theta = \lambda_{\text{st}}.$$

We know the pairs of coordinates relate to one another via following four relations:

$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \\ p_r = \frac{x}{r}p_x + \frac{y}{r}p_y \\ p_\theta = -yp_x + xp_y \end{cases}$$

Differentiating the first two relations using the chain- and product rule yields:

$$\begin{cases} dr = \frac{xdx+yd y}{r} \\ d\theta = \frac{\frac{1}{x}dy - \frac{y}{x^2}dx}{1+(\frac{y}{x})^2} = \frac{\frac{1}{x^2}(xdy-ydx)}{\frac{x^2+y^2}{x^2}} = \frac{xdy-ydx}{r^2} \end{cases}$$

Using this we may calculate:

$$\begin{aligned} p_r dr + p_\theta d\theta &= \left(\frac{x}{r}p_x + \frac{y}{r}p_y \right) \cdot \left(\frac{xdx+yd y}{r} \right) + (-yp_x + xp_y) \cdot \left(\frac{xdy-ydx}{r^2} \right) \\ &= dx \left(\frac{x^2p_x + xyp_y - xyp_y + y^2p_x}{r^2} \right) + dy \left(\frac{y^2p_y + xyp_x - xyp_x + x^2p_y}{r^2} \right) \\ &= p_x dx + p_y dy = \lambda_{st}. \end{aligned}$$

Thus, we have shown that coordinates $(r, p_r, \theta, p_\theta)$ form a Darboux chart of the open set $\{y \neq 0\}$. $\square$

1.3 (c)

Suppose we are given a potential function $V: \mathbb{R}^2 \rightarrow \mathbb{R}$ which assigns to every position $q := (x, y)$ a potential $V(q)$. Then, we may define the total energy function $H(p, q)$ as

$$H: \mathbb{R}^4 \rightarrow \mathbb{R}, (x, p_x, y, p_y) \mapsto H(p, q) := \frac{\|p\|^2}{2m} + V(q),$$

where $p = (p_x, p_y)$ are the momentum variables and m is some positive constant. Suppose in the following also, that V is rotationally symmetric: $\partial_\theta V \equiv 0$.

We will rewrite H in the coordinates of part (b) to show that is invariant under the diffeomorphism family given by $\varphi_t(r, p_r, \theta, p_\theta) = (r, p_r, \theta + t, p_\theta)$. Let us avert our attention to the term $\|p\|^2$ first. We know that the following two relations hold:

$$\begin{cases} p_r = \frac{x}{r}p_x + \frac{y}{r}p_y \\ p_\theta = -yp_x + xp_y \end{cases}$$

From this, we may deduce

$$p_r^2 \cdot r^2 + p_\theta^2 = p_x^2(x^2 + 2xyp_xp_y - 2xyp_xp_y + y^2) + p_y^2(x^2 + 2xyp_xp_y - 2xyp_xp_y + y^2) = p_x^2r^2 + p_y^2r^2.$$

Equivalently, we find that $p_x^2 + p_y^2 = p_r^2 + \frac{p_\theta^2}{r^2}$. Hence, we may rewrite the total energy function as

$$H(r, p_r, \theta, p_\theta) = \frac{p_r^2 + \frac{p_\theta^2}{r^2}}{2m} + V(x(r, \theta), y(r, \theta))$$

Now clearly the first summand is invariant under every diffeomorphism φ_t and by the assumption on V , the second term is as well. We conclude that H is invariant under rotations. We will now show that the pullback of w_{st} remains invariant under these diffeomorphisms as well. Indeed:

$$\begin{aligned} \varphi_t^* w_{\text{st}} &= \varphi_t^* (dr \wedge dp_r + d\theta \wedge dp_\theta) \\ &= \varphi_t^* (dr) \wedge \varphi_t^* (dp_r) + \varphi_t^* (d\theta) \wedge \varphi_t^* (dp_\theta) \\ &= dr \wedge dp_r + d(\theta + t) \wedge dp_\theta \\ &= dr \wedge dp_r + d\theta \wedge dp_\theta \\ &= w_{\text{st}}. \end{aligned}$$

We may notice that the diffeomorphisms φ_t are the flowlines of the vector field ∂_θ (extended via continuity to all of $\mathbb{R}^4$). We may also notice that $\mathbb{R}^4$ has trivial De-Rahm cohomology since it is contractible. Thus, problem (a) guarantees that ∂_θ is the Hamiltonian vector-field of a smooth function $F: \mathbb{R}^4 \rightarrow \mathbb{R}$, which is invariant under φ_t . We will deduce a formula which expresses F in the coordinates $(r, p_r, \theta, p_\theta)$. By the definition of F , we have

$$w_{\text{st}}(\partial_\theta, \cdot) = -dF.$$

We may also write $dF = \frac{\partial F}{\partial r} dr + \frac{\partial F}{\partial p_r} dp_r + \frac{\partial F}{\partial \theta} d\theta + \frac{\partial F}{\partial p_\theta} dp_\theta$. This lets us deduce the following system of equations

$$\begin{cases} w_{\text{st}}(\partial_\theta, \partial_r) = -\frac{\partial F}{\partial r} \\ w_{\text{st}}(\partial_\theta, \partial_{p_r}) = -\frac{\partial F}{\partial p_r} \\ w_{\text{st}}(\partial_\theta, \partial_\theta) = -\frac{\partial F}{\partial \theta} \\ w_{\text{st}}(\partial_\theta, \partial_{p_\theta}) = -\frac{\partial F}{\partial p_\theta}. \end{cases}$$

Plugging in the polar-coordinate representation of w_{st} which we deduced in (b), we find that

$$\begin{cases} \frac{\partial F}{\partial r} = 0 \\ \frac{\partial F}{\partial p_r} = 0 \\ \frac{\partial F}{\partial \theta} = 0 \\ \frac{\partial F}{\partial p_\theta} = 1. \end{cases}$$

The solution space to this is $p_\theta + c$, where c is an arbitrary constant. Thus, $F = p_\theta + c$ for some $c \in \mathbb{R}$. If we prefer to write it in standard coordinates, it is given by $-yp_x + xp_y + c$.

2 Problem 2

Let M be a smooth manifold and ∇ be a symmetric connection on the tangent bundle of M . Furthermore, equip the tensor bundles $\Gamma_k^k(M) \rightarrow M$ with the connections induced by ∇ . Throughout this problem let $X, Y, Z \in \mathfrak{X}(M)$ be smooth vector-fields on M ,

2.1 (a)

Prove that

$$d\lambda(X, Y) = (\nabla_X \lambda)Y - (\nabla_Y \lambda)X.$$

Proof. Using Proposition 8.7 we may write:

$$d\lambda(X, Y) = \mathcal{L}_X(\lambda(Y)) - \mathcal{L}_Y(\lambda(X)) - \lambda[X, Y].$$

Due to (21.5) in the lecture notes, we may rewrite the appearing Lie-derivatives in terms of the covariant derivative:

$$\begin{aligned}\mathcal{L}_X(\lambda(Y)) &= (\nabla_X \lambda)(Y) + \lambda(\nabla_X(Y)) \\ \mathcal{L}_Y(\lambda(X)) &= (\nabla_Y \lambda)(X) + \lambda(\nabla_Y(X)).\end{aligned}$$

Plugging these relations into the original equation, and using the fact that ∇ is symmetric, we obtain:

$$\begin{aligned}d\lambda(X, Y) &= (\nabla_X \lambda)(Y) + \lambda(\nabla_X(Y)) - ((\nabla_Y \lambda)(X) + \lambda(\nabla_Y(X))) - \lambda[X, Y] \\ &= (\nabla_X \lambda)(Y) - (\nabla_Y \lambda)(X) + \lambda(T(X, Y)) \\ &= (\nabla_X \lambda)(Y) - (\nabla_Y \lambda)(X).\end{aligned}$$

□

2.2 (b)

Prove that

$$(\mathcal{L}_X \lambda)(Y) = (\nabla_X \lambda)(Y) + \lambda(\nabla_Y(X))$$

Proof. Using Cartan's magic formula, we obtain

$$(\mathcal{L}_X \lambda)(Y) = d(\lambda(X))(Y) + (d\lambda)(X, Y).$$

Using the definition of the Lie-derivative of smooth functions, we may rewrite the first summand as $d(\lambda(X))(Y) = \mathcal{L}_Y(\lambda(X))$. Once again using the identity (21.5) from the lecture notes we have $\mathcal{L}_Y(\lambda(X)) = (\nabla_Y \lambda)(X) + \lambda(\nabla_Y X)$. Combining this with the equation deduced in (a), we therefore obtain:

$$\begin{aligned}(\mathcal{L}_X \lambda)(Y) &= (\nabla_Y \lambda)(X) + \lambda(\nabla_Y X) + (d\lambda)(X, Y) \\ &= (\nabla_Y \lambda)(X) + \lambda(\nabla_Y X) + (\nabla_X \lambda)(Y) - (\nabla_Y \lambda)(X) \\ &= (\nabla_X \lambda)(Y) + \lambda(\nabla_Y(X)).\end{aligned}$$

□

2.3 (c)

Let $S \in \Gamma(T_2^0 M)$. Prove that

$$(\mathcal{L}_X S)(X, Y) = (\nabla_X S)(Y, Z) + S(\nabla_Y X, Z) + S(Y, \nabla_Z X).$$

Proof. Clearly the value of $\mathcal{L}_X S(X, Y)$ at a point P , depends only on an open neighborhood of P . In particular we may pass to a sufficiently small neighborhood U of P which is the domain of a chart $x: U \rightarrow \mathbb{R}^n$. This chart gives rise to a frame $\partial_1, \dots, \partial_n$ of the tangent bundle, where $\partial_i = x^* e_i$. Dualizing these vectors, we arrive at a frame $\partial_1^*, \dots, \partial_n^*$ of the dual bundle. Finally, we arrive at a frame of $\Gamma(T_2^0 M)$ given by vectors of the form $\partial_i^* \otimes \partial_j^*$, for $i, j = 1, \dots, n$. In particular, we may express S in these coordinates as

$$S = S^{i,j}(\partial_i^* \otimes \partial_j^*).$$

This implies that locally S may be written as a sum of elements which have the form $\lambda \otimes \mu$ where $\lambda, \mu \in \Omega^1(M)$. If we knew the statement to be true for such elements, then $\mathbb{R}$ -linearity of $\mathcal{L}_X$ and ∇_X would complete the proof. Thus, it suffices to prove the claim for when $S = \lambda \otimes \mu$ with $\lambda, \mu \in \Omega^1(M)$. Before we do this, let us quickly derive a Leibnitz rule for the Lie-derivative of a tensor product. Suppose we have 1-forms $\lambda, \mu \in \Omega^1(M)$ and a vector-field $X \in \mathfrak{X}(M)$. We would like to evaluate $\mathcal{L}_X(\lambda \otimes \mu)$:

$$\begin{aligned} \mathcal{L}_X(\lambda \otimes \mu) &= \frac{d}{dt} \left((\varphi_X^t)^*(\lambda \otimes \mu) \right) \Big|_{t=0} \\ &= \frac{d}{dt} \left((\varphi_X^t)^*(\lambda) \otimes (\varphi_X^t)^*(\mu) \right) \Big|_{t=0}. \end{aligned}$$

If we define $F(t, s) := (\varphi_X^t)^*(\lambda) \otimes (\varphi_X^s)^*(\mu)$, the product rule ensures :

$$\frac{d}{dt} F(t, t) \Big|_{t=0} = \left(\frac{d}{dt} (\varphi_X^t)^* \lambda \Big|_{t=0} \right) \otimes \mu + \lambda \otimes \left(\frac{d}{dt} (\varphi_X^t)^* \mu \Big|_{t=0} \right) = (\mathcal{L}_X \lambda) \otimes \mu + \lambda \otimes \mathcal{L}_X(\mu),$$

which is the Leibnitz rule that we wanted to prove. Using this Leibnitz rule, we can complete the proof in the following calculation:

$$\begin{aligned} \mathcal{L}_X(\lambda \otimes \mu)(Y, Z) &= ((\mathcal{L}_X \lambda) \otimes \mu)(Y, Z) + (\lambda \otimes \mathcal{L}_X \mu)(Y, Z) \\ &= (\mathcal{L}_X \lambda)(Y) \cdot \mu(Z) + \lambda(Y) \cdot (\mathcal{L}_X \mu)(Z) \\ &\stackrel{(b)}{=} ((\nabla_X \lambda)(Y) + \lambda(\nabla_Y X)) \cdot \mu(Z) + \lambda(Y) \cdot ((\nabla_X \mu)(Z) + \mu(\nabla_Z Y)) \\ &= ((\nabla_X \lambda)(Y) \cdot \mu(Z) + \lambda(Y) \cdot (\nabla_X \mu)(Z)) + (\lambda(\nabla_Y X) \cdot \mu(Z)) + (\lambda(Y) \cdot \mu(\nabla_Z Y)) \\ &= (\nabla_X(\mu \otimes \lambda))(Y, Z) + (\mu \otimes \lambda)(\nabla_Y X, Z) + (\mu \otimes \lambda)(Y, \nabla_Z X), \end{aligned}$$

which is exactly what we wanted to prove. $\square$

2.4 (d)

Assume ∇ is the Levi-Civita connection for a pseudo-Riemannian manifold (M, g) and denote $\langle \cdot, \cdot \rangle := g$. Show that for a vector field $X \in \mathfrak{X}(M)$, the corresponding tensor field $\nabla X_{\flat} \in \Gamma(T_2^0 M)$ is anti-symmetric if and only if X satisfies the killing equation:

$$\langle \nabla_Y X, Z \rangle + \langle Y, \nabla_Z X \rangle \equiv 0 \quad \text{for all } X, Y \in \mathfrak{X}(M) .$$

Proof. Let us first recall the definition of the tensor $(\nabla X)_{\flat}$. We have defined it such that for vector-fields $X, Y \in \mathfrak{X}(M)$ we have

$$(\nabla X_{\flat})(Y, Z) = (\nabla_Y X)_{\flat}(Z).$$

Thus, we are interested in $(\nabla_Y X)_{\flat}(Z) + (\nabla_Z X)_{\flat}(Y)$ vanishing.

Our first claim is that the musical-isomorphism $\flat$ commutes with the co-variant derivative. We will show this by calculating the same quantity in two different ways. Firstly, since ∇ is a metric connection, exercise 22.3 in the lecture notes guarantees

$$\mathcal{L}_Y(\langle X, Z \rangle) = \langle \nabla_Y X, Z \rangle + \langle X, \nabla_Y Z \rangle.$$

On the other hand, we may use the Leibnitz rule (equation 21.5 in the lecture notes) to evaluate this as:

$$\mathcal{L}_Y(\langle X, Z \rangle) = \mathcal{L}_Y(X_{\flat}(Z)) = (\nabla_Y X)_{\flat}(Z) + X_{\flat}(\nabla_Y Z) = (\nabla_Y X)_{\flat}(Z) + \langle X, \nabla_Y Z \rangle$$

Equating the last term in the above equation to the previously calculated term, we see that $\nabla_Y(X_{\flat}) = (\nabla_Y X)_{\flat}$. Thus, we may calculate:

$$\begin{aligned} (\nabla_Y X)_{\flat}(Z) + (\nabla_Z X)_{\flat}(Y) &= (\nabla_Y X)_{\flat}(Z) + (\nabla_Z X)_{\flat}(Y) \\ &= \langle \nabla_Y X, Z \rangle + \langle \nabla_Z X, Y \rangle \\ &= \langle \nabla_Y X, Z \rangle + \langle Y, \nabla_Z X, \rangle \end{aligned}$$

Thus, we have shown, that the killing equation is the same equation as the anti-symmetry equation, simply rewritten in other symbols. $\square$

2.5 (e)

Assume M is compact, and $X \in \mathfrak{X}(M)$ is a vector field which satisfies the killing equation. Then, X admits global flow and the diffeomorphism family $\{\varphi_X^t\}_{t \in \mathbb{R}}$ is metric-preserving, i.e. they are isometries.

Proof. If M is compact then clearly X has compact support and thus theorem 5.1 guarantees that X admits global flow. We are left to show that the diffeomorphisms φ_X^t are isometries: $(\varphi_X^t)^* g = g$. Equivalently, we can show that

$$\mathcal{L}_X(g) \equiv 0.$$

Let $X, Z \in \mathfrak{X}(M)$, since g is a type $(0, 2)$ tensor field, we may apply (c) to calculate:

$$\mathcal{L}_X(g)(Y, Z) = (\nabla_X g)(Y, Z) + (\langle \nabla_Y X, Z \rangle + \langle Y, \nabla_Z X \rangle).$$

Since ∇ is a metric connection $\nabla g \equiv 0$ (exercise 22.3) and thus the first summand vanishes. Since X is antisymmetric, (d) ensures that the second summand vanishes as well. $\square$

3 Problem 3

Let $(\mathbb{H}, h)$ be the Poincaré half-plane as described in the the problem set. We will investigate geodesics on this manifold.

3.1 (a)

A smooth path $\gamma(t) = (x(t), y(t))$ is a geodesic on $\mathbb{H}$ if and only if its coordinates satisfy the following system of second-order ordinary differential equations:

$$\begin{aligned} \ddot{x} - \frac{2}{y} \dot{x} \dot{y} &= 0 \\ \ddot{y} + \frac{1}{y} (\dot{x}^2 - \dot{y}^2) &= 0. \end{aligned}$$

Proof. Let x and y denote the standard coordinates of $\mathbb{H}$ with corresponding vector fields ∂_x and ∂_y . Furthermore, if ∇ is the induced Levi-Civita connection on $\mathbb{H}$ by h , equation 22.1 in the lecture notes ensures that γ is a geodesic if and only if its coordinates solve the following second order system of differential equations:

$$\begin{aligned} \ddot{x} + \Gamma_{xx}^x \dot{x}^2 + (\Gamma_{xy}^x + \Gamma_{yx}^x) \dot{x} \dot{y} + \Gamma_{yy}^x \dot{y}^2 &= 0 \\ \ddot{y} + \Gamma_{xx}^y \dot{x}^2 + (\Gamma_{xy}^y + \Gamma_{yx}^y) \dot{x} \dot{y} + \Gamma_{yy}^y \dot{y}^2 &= 0. \end{aligned}$$

To avoid confusion, let us clarify the above equations. Here the x and y appearing in the Christoffel symbols refer to the coordinates vectors of the chart, Contrary to this, the terms $\dot{x}, \ddot{x}, \dot{y}, \ddot{y}$, refer to the component functions of the path γ .

Since the Levi-Civita connection is symmetric, the above equations simplify to

$$\ddot{x} + \Gamma_{xx}^x \dot{x}^2 + 2\Gamma_{xy}^x \dot{x} \dot{y} + \Gamma_{yy}^x \dot{y}^2 = 0 \quad (1)$$

$$\ddot{y} + \Gamma_{xx}^y \dot{x}^2 + 2\Gamma_{xy}^y \dot{x} \dot{y} + \Gamma_{yy}^y \dot{y}^2 = 0. \quad (2)$$

We are left to compute the Christoffel-Symbols. In standard coordinates, we may represent any two vector fields $X, Y \in \mathfrak{X}(\mathbb{H})$ as $X = X^1 \partial_x + X^2 \partial_y$ and $Y = Y^1 \partial_x + Y^2 \partial_y$. Thus, in these coordinates, we may express h in matrix form:

$$h_{(x,y)}(X, Y) = \begin{bmatrix} Y^1 \partial_x & Y^2 \partial_y \end{bmatrix} \begin{bmatrix} \frac{1}{y^2} & 0 \\ 0 & \frac{1}{y^2} \end{bmatrix} \begin{bmatrix} X^1 \partial_x \\ X^2 \partial_y \end{bmatrix}.$$

Let us denote by $h_{i,j}$ the entries of the above matrix. Next, let us push-forward h to $\Omega^1(M)$ via the musical bundle isomorphism $\flat$. To do this, note that $(\partial_x)_\flat(\partial_x) = h(\partial_x, \partial_x) = \frac{1}{y^2}$ and $(\partial_x)_\flat(\partial_y) = h(\partial_x, \partial_y) = 0$, where y is the second coordinate of the projection $\pi(\partial_x)$. In particular, we find that $(\partial_x)_\flat = \frac{1}{y^2}dx$. Completely analogously, we find that $(\partial_y)_\flat = \frac{1}{y^2}dy$. These two equations uniquely determine the inverse musical isomorphism $\sharp$, namely $(dx)^\sharp = y^2\partial_x$ and $(dy)^\sharp = y^2\partial_y$. Similarly to before, this yields a matrix representation of $(\flat)_*h$ as

$$\begin{bmatrix} h(dx^\sharp, dx^\sharp) & h(dx^\sharp, dy^\sharp) \\ h(dy^\sharp, dx^\sharp) & h(dy^\sharp, dy^\sharp) \end{bmatrix} = \begin{bmatrix} h(y^2\partial_x, y^2\partial_x) & 0 \\ 0 & h(y^2\partial_y, y^2\partial_y) \end{bmatrix} = \begin{bmatrix} y^2 & 0 \\ 0 & y^2 \end{bmatrix}.$$

Let us denote the entries of this matrix as $h^{i,j}$ and define $\partial_1 = \partial_x$ and $\partial_2 = \partial_y$. Due to equation 22.6 in the lecture notes, we may relate the Christoffel symbols to the entries $h_{i,j}$ and $h^{k,l}$ as follows:

$$T_{x,x}^x = \frac{1}{2}h^{k,1}(\partial_x(h_{1,k}) + \partial_x(h_{1,k}) - \partial_k(h_{1,1})) \quad (\text{Einstein convention}).$$

Noticing that $h_{1,k}$ does not depend on x , the first two summands in the bracket vanish. Thus, the above simplifies to

$$T_{x,x}^x = -\frac{1}{2}h^{k,1}(\partial_k(h_{1,1})) = 0$$

Where we used the fact that the right factor vanishes for $k = 1$ and the left for $k = 2$. Similarly let us compute the other symbols

$$\begin{aligned} T_{x,y}^x &= \frac{1}{2}h^{k,1}(\partial_1(h_{2,k}) + \partial_2(h_{1,k}) - \partial_k(h_{1,2})) \\ &= \frac{1}{2}h^{k,1}(\partial_2(h_{1,k}) - \partial_k(h_{1,2})) \\ &= \frac{1}{2}h^{1,1}(\partial_2(h_{1,1}) - \partial_1(h_{1,2})) \\ &= \frac{1}{2}y^2 \cdot \left(-\frac{2}{y^3}\right) = -\frac{1}{y}. \end{aligned}$$

and,

$$\begin{aligned} T_{y,y}^x &= \frac{1}{2}h^{k,1}(\partial_2(h_{2,k}) + \partial_2(h_{2,k}) - \partial_k(h_{2,2})) \\ &= \frac{1}{2}h^{1,1}(\partial_2(h_{2,1}) + \partial_2(h_{2,1}) - \partial_1(h_{2,2})) \\ &= 0 \end{aligned}$$

where we used that $h_{i,j}$ vanishes when the indices don't match and it doesn't depend on x . $\square$

We repeat this procedure three more times:

$$\begin{aligned}
T_{y,y}^y &= \frac{1}{2}h^{k,2}(\partial_2(h_{2,k}) + \partial_2(h_{2,k}) - \partial_k(h_{2,2})) \\
&= \frac{1}{2}h^{2,2}(\partial_2(h_{2,2}) + \partial_2(h_{2,2}) - \partial_2(h_{2,2})) \\
&= \frac{1}{2}y^2 \cdot (-2\frac{1}{y^3}) \\
&= -\frac{1}{y}.
\end{aligned}$$

$$\begin{aligned}
T_{x,x}^y &= \frac{1}{2}h^{k,2}(\partial_1(h_{1,k}) + \partial_1(h_{1,k}) - \partial_k(h_{1,1})) \\
&= \frac{1}{2}h^{2,2}(\partial_1(h_{1,2}) + \partial_1(h_{1,2}) - \partial_2(h_{1,1})) \\
&= \frac{1}{2}h^{2,2}(-\partial_2(h_{1,1})) \\
&= \frac{1}{y}.
\end{aligned}$$

$$\begin{aligned}
T_{x,y}^y &= \frac{1}{2}h^{k,2}(\partial_1(h_{2,k}) + \partial_2(h_{1,k}) - \partial_k(h_{1,2})) \\
&= \frac{1}{2}h^{2,2}(\partial_1(h_{2,2}) + \partial_2(h_{1,2}) - \partial_2(h_{1,2})) \\
&= \frac{1}{2}h^{2,2}(-\partial_2(h_{1,2})) \\
&= 0.
\end{aligned}$$

Plugging the calculated Christoffel symbols into equations (1) and (2), we obtain:

$$\begin{aligned}
\ddot{x} + \Gamma_{xx}^x \dot{x}^2 + 2\Gamma_{xy}^x \dot{x}\dot{y} + \Gamma_{yy}^x \dot{y}^2 &= \ddot{x} - \frac{2}{y}\dot{x}\dot{y} \\
\ddot{y} + \Gamma_{xx}^y \dot{x}^2 + 2\Gamma_{xy}^y \dot{x}\dot{y} + \Gamma_{yy}^y \dot{y}^2 &= \ddot{y} + \frac{1}{y}\dot{x}^2 - \frac{1}{y}\dot{y}^2.
\end{aligned}$$

Since these are exactly the equations which we claimed to be the geodesic equations we have completed the proof.

3.2 (b)

let x_0 in $\mathbb{R}$ and $r > 0$. Show that the above geodesic equations admit solutions of the form

$$\begin{aligned}
(x(t), y(t)) &= (x_0, y(t)) \quad (\text{for some function } y(t) > 0) \text{ and} \\
(x(t), y(t)) &= (x_0 + r \cos \theta(t), r \sin \theta(t)) \quad (\text{for some function } \theta(t) \in (0, \pi)).
\end{aligned}$$

Proof. Let us consider the first class of solutions first. If we assume the x -coordinate to be constant, the geodesic equations reduce to:

$$\ddot{y} = \frac{1}{y} \dot{y}^2.$$

we may easily verify that the family $\{y(t) = \lambda \exp(\mu t) \mid \lambda, \mu \in \mathbb{R},\}$ solves this system. Here λ and μ uniquely determine starting velocity and starting y -coordinate. Now let us continue to the second class of solutions. Assume $(x(t), y(t)) = (x_0 + r \cos \theta(t), r \sin \theta(t))$. This transforms the geodesic equations into:

$$\begin{aligned} \ddot{\theta}(t) \sin(\theta(t)) &= \cos(\theta(t)) \dot{\theta}(t)^2. \\ \ddot{\theta}(t) &= \cot(\theta(t)) \cdot \dot{\theta}(t)^2. \end{aligned}$$

By the Picard-Lindelöf theorem, the second equation has solutions for any starting value $\theta(t_0)$ and starting velocity $\dot{\theta}(t)$. Now, we may notice that any such a solution is also a solution to the first equation since the first equation is simply the second equation multiplied by $\sin(\theta(t))$. Thus, we always have solutions of the desired form, at least locally.

If we prefer, we can be more explicit and consult an ODE solver, which produces the solution set $\theta(t) = 2 \arctan(e^{c_1 t + c_2}) \in (0, \pi)$. $\square$

3.3 (c)

We will give a heuristic argument for why we believe the geodesics appearing in (b) are the only ones on $\mathbb{H}$.

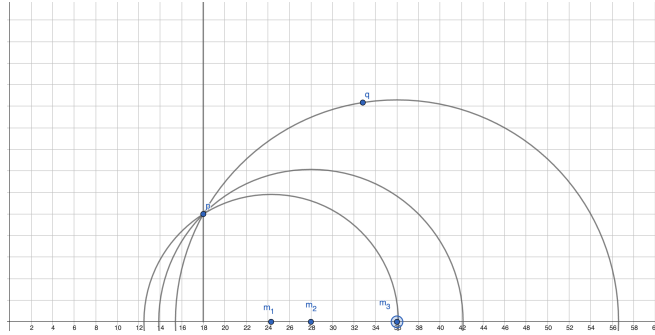


Figure 1: The Poincaré Half-Plane with Geodesics

Let p be any point on $\mathbb{H}$. As we have seen in part (b) we have an infinite family of geodesics through p , namely one vertical line and the collection of half-circles which have midpoint on the x -axis and pass through p . Now geodesics are uniquely determined by two conditions: their starting velocity and their starting point. From the figure it is visually clear that for any starting velocity, there

is a unique circle from our collection (or line if its pointing upward/downward) which has this velocity vector in its tangent space at p . Reparameterizing this geodesic if needed, we can assume that it has exactly this starting velocity at p . Since geodesics are uniquely determined by starting point and starting velocity, every geodesic corresponds to one from our collection.

3.4 (d)

Prove that $\exp(Y)$ is defined for all $Y \in T\mathbb{H}$ which point upward or downwards.

Proof. Let $p = (p_1, p_2) \in \mathbb{H}$ be any point and $Y = \lambda \cdot \partial_y$ be a tangent vector at p pointing vertical. If we knew that the geodesic at p given by γ_{∂_y} (notation from the script) were defined for all $t \in \mathbb{R}$, we would obtain that $\exp(Y)$ is defined as well. Indeed, this would suffice since we could reparametrize $\gamma_{\partial_y}(t)$ as $\gamma_Y(t) := \gamma_{\partial_y}(\lambda t)$, such that it has starting velocity Y instead of ∂_y and then notice that $\gamma_Y(1) = \gamma_{\partial_y}(\lambda)$. Thus, it suffices to show that $\gamma_{\partial_y}(t)$ is defined for all $t \in \mathbb{R}$. However, this geodesic is the unique geodesic which starts at p with starting velocity ∂_y and we have already calculated it in (b). Namely:

$$\gamma_{\partial_y}: \mathbb{R} \rightarrow \mathbb{H}, t \mapsto (p_1, p_2 \exp(\frac{t}{p_1})),$$

which was globally defined. □